

mlf qz 1

Question 1 : 6406532158886

View Solutions (0) Total Mark : 3.00 | Type : MCQ

What is the best linear approximation of

$$f(x, y) = \sqrt{x + y}$$

around

$$(4, 5)?$$

Options :

$$L(x, y) = 9 + \frac{1}{6}(x - 5) + \frac{1}{6}(y - 4)$$

$$L(x, y) = 3 + \frac{1}{6}(x - 4) + \frac{1}{6}(y - 5)$$

$$L(x, y) = 3 + \frac{1}{3}(x - 4) + \frac{1}{3}(y - 5)$$

$$L(x, y) = 9 + \frac{1}{18}(x - 4) + \frac{1}{18}(y - 5)$$

Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 2 : 6406532158887

View Solutions (0) Total Mark : 3.00 | Type : MCQ

A company manufactures a special rectangular metal sheet. The production cost (in hundreds of rupees) depends on its length

x

meters and width

y

meters according to

$$C(x, y) = x^2 + y^2$$

. At a particular stage, the dimensions of the sheet are

$(1, 2)$

. The company plans to change the dimensions in the direction represented by the vector

$$v = \frac{1}{5}(3, 4)$$

and wants to determine how rapidly the production cost changes in that specific direction. Find the directional derivative of

$C(x, y)$

at the point

$(1, 2)$

in the direction of

$$v$$

.

Options :

$$5$$

$$4$$

$$\frac{11}{5}$$

$$\frac{22}{5}$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 3 : 6406532158888

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Find the projection matrix that projects the

$$xy$$

plane onto the line

$$x + y = 0$$

.

Options :

$$\frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 4 : 6406532158889

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Let

$$M$$

be a real symmetric matrix. Consider the following statements. (1)

$$M$$

is orthogonally diagonalizable. (2)

$$M$$

is not diagonalizable. (3) All the eigenvalues of

$$M$$

are real. (4) The eigenvalues of

$$M$$

may be imaginary. (5) Eigenvectors corresponding to distinct eigenvalues of

$$M$$

are linearly independent. (6) Eigenvectors corresponding to distinct eigenvalues of

$$M$$

are linearly dependent. Which of the above statements are true?

Options :

(1), (3), (5)

(2), (4), (6)

(2), (3), (5)

(1), (4), (6)

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 5 : 6406532158890

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Let

$$A$$

and

$$B$$

be two square matrices of order

4

. The rows of

B

are represented by row vectors

r_1, r_2, r_3, r_4

, from top to bottom, and the columns of

B

are represented by column vectors

c_1, c_2, c_3, c_4

, from left to right. The second row of

A

is

$[1, 3, -2, 4]$

. Which of the following is/are true?

Options :

☐ The second row of is .

$$AB$$

$$r_1 + 3r_2 - 2r_3 + 4r_4$$

☐ The second column of is .

$$AB$$

$$c_1 + 3c_2 - 2c_3 + 4c_4$$

☐ The row space of is contained in the row space of .

$$AB$$

$$A$$

☐ The row space of is contained in the row space of .

$$AB$$

$$B$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

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Question 6 : 6406532158891

View Solutions (0) Total Mark : 3.00 | Type : MSQ

$$P$$

is a square matrix of order

$$2$$

with the eigenvalue

$$3$$

repeated twice. Which of the following is/are true?

Options :

☐ is invertible.

$$P$$

☐ Each of the entries on the main diagonal of is equal to .

$$P$$

$$3$$

☐ has two linearly independent eigenvectors.

$$P$$

☐ is not invertible, where is the identity matrix.

$$P - 3I$$

$$I$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 7 : 6406532158892

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Suppose

$$M$$

is a

$$6 \times 4$$

matrix with rank 4. Which of the following is/are true?

Options :

☐ Columns of M are linearly independent.

$$M$$

☐ Null space of M contains only the zero vector.

$$M$$

☐ Row space of M is $\mathbb{R}^6$.

$$M$$

$$\mathbb{R}^6$$

☐ Left null space of M has dimension 2.

$$M$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 8 : 6406532158893

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Which of the following represents the set of all points on the line passing through

$$(2, 1)$$

along the direction

$$(-3, 2)$$

? (More than one option can be correct).

Options :

☐

$$\{(2, 1) + \alpha(-3, 2) : \alpha \in \mathbb{R}\}$$

☐

$$\{(x, y) : 2x + 3y = 7; x, y \in \mathbb{R}\}$$

☐

$$\{(-3, 2) + \alpha(2, 1) : \alpha \in \mathbb{R}\}$$

☐

$$\{(x, y) : 2y - x = 7; x, y \in \mathbb{R}\}$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 9 : 6406532158894

View Solutions (0) Total Mark : 3.00 | Type : SA

Let

$$u, v \in \mathbb{R}^5$$

be two vectors with

$$\|u\| = 2$$

and

$$\|v\| = 3$$

. Find the maximum possible value of

$$|u^T v|$$

, where

$$|x|$$

represents the absolute value of

$$x$$

. Your answer should be an integer.

Answer (Numeric):

Accepted Answer : 6

Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 10 : 6406532158895

View Solutions (0) Total Mark : 4.00 | Type : SA

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be the matrix whose eigenvalues are

$$1$$

and

$$4$$

, and whose eigenvectors are

$$(3, 1)$$

and

$$(2, 1)$$

respectively. Find

$$a + b + c + d$$

. Your answer should be an integer.

Answer (Numeric):

Accepted Answer : 20

Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 11 : 6406532158896

Total Mark : 0.00 | Type : COMPREHENSION

Consider the following dataset consisting of two features

$$x_1$$

and

$$x_2$$

:

x_1	x_2
2	4
6	8

We want to reduce the dimensionality of the dataset from

$$2$$

to

$$1$$

using the following encoder- decoder pairs.

$$\text{Pair 1: } f(x_1, x_2) = \frac{x_1 + x_2}{2}, \quad g(u) = [u, u]$$

$$\text{Pair 2: } f(x_1, x_2) = x_1 - x_2, \quad g(u) = [u, u]$$

The reconstruction loss is defined as the mean squared distance between the reconstructed input and the original input. Based on the above information, answer the given sub-questions.

Your score : 0

Broken Format?

Question 12 : 6406532158897

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : SA

Compute the reconstruction loss for Pair 1. NOTE: Your answer should be an integer

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 13 : 6406532158898

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : SA

Compute the reconstruction loss for Pair 2. NOTE: Your answer should be an integer

Answer (Numeric):

Accepted Answer : 108

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 14 : 6406532158899

View Parent QNView Solutions (0) Total Mark : 1.00 | Type : MCQ

Which encoder-decoder pair performs better?

Options :

Pair-1

Pair-2

Your score : 0

Question 1 : 6406531951381

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Let the vectors

$$u = (1, a, 2), v = (2, -1, b)$$

be orthogonal. Which of the following must be true?

Options :

$$a - 2b = 2$$

$$a + 2 = 2b$$

$$2b - a = 2$$

$$a - 2b = 0$$

Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 2 : 6406531951385

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Consider the function

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

:

$$f(x) = \begin{cases} x^2 + 2, & x \leq 1 \\ x + 2, & x > 1 \end{cases}$$

Which of the following is true?

Options :

is continuous at , but not differentiable at .

$$f$$

$$x = 1$$

$$x = 1$$

is both continuous and differentiable at .

$$f$$

$$x = 1$$

is neither continuous nor differentiable at .

$$f$$

$$x = 1$$

is differentiable at , but not continuous at .

$$f$$

$$x = 1$$

$$x = 1$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 3 : 6406531951382

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following matrices:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Which of the following is true?

Options :

is diagonalizable, is not diagonalizable.

A

B

is not diagonalizable, is diagonalizable.

A

B

Both *A* and *B* are diagonalizable.

A

B

Neither A nor B is diagonalizable

A

B

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 4 : 6406531951383

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider a dataset

D

for a regression problem where each element is of the form

(x_i, y_i)

, for

$i = 1, 2, 3$

.

x_i

is the feature and

$$y_i$$

is the label of the

$$i^{th}$$

data-point.

$$D = \{(1, 1), (2, 4), (3, 7)\}$$

Let

$$f_1, f_2 : \mathbb{R} \rightarrow \mathbb{R}$$

be two regression models such that:

$$f_1(x) = 2x - 1, \quad f_2(x) = 3x - 2$$

Which of the following options is correct?

Options :

has the lower mean squared loss.

$$f_1(x)$$

has the lower mean squared loss.

$$f_2(x)$$

Both models have same mean squared loss.

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 5 : 6406531951384

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Which of the following is a basis for the column space of the matrix

A

given below?

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Options :

$$\{(1, 2, 1), (1, -1, 0)\}$$

$$\{(1, 0, 0), (0, 1, 0)\}$$

$$\{(1, 2, 1), (1, -1, 0), (2, 1, 1)\}$$

$$\{(1, 3, 2), (1, -1, 0)\}$$

Your score : 0

Discussions (1) [Suggest Correction](#) [Broken Format?](#)

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Question 6 : 6406531951386

View Solutions (0) Total Mark : 3.00 | Type : MSQ

The matrix

$$P = \begin{bmatrix} m & 2 \\ 1 & n \end{bmatrix}$$

has eigenvalues

$$0$$

and

$$3$$

. Which of the following options is/are correct?

Options :

☐

$$mn = 0$$

☐

$$mn = 2$$

☐

$$m + n = 3$$

☐

$$m + n = 2$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 7 : 6406531951387

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following is/are true?

Options :

☐ If A and B are matrices of the same size, then $\text{rank}(A + B) = \text{rank}(A) + \text{rank}(B)$.

A

B

$$\text{rank}(A + B) = \text{rank}(A) + \text{rank}(B)$$

☐ The rank of a matrix is equal to the number of non-zero rows in it.

☐ If A is a matrix which is obtained by permuting the rows of the 5×5 identity matrix, then the rank of A is 5 .

A

5×5

A

5

☐ If u is a non-zero column vector in $\mathbb{R}^{10}$, then the rank of the matrix uu^T is 1 .

u

$\mathbb{R}^{10}$

uu^T

1

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 8 : 6406531951388

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Let

$$L_1$$

and

$$L_2$$

be two lines in

$$\mathbb{R}^3$$

defined in parametric form:

$$L_1 : \quad x = 1 + 4t, \quad y = 5 - 4t, \quad z = -1 + 5t, \quad t \in \mathbb{R}$$

$$L_2 : \quad x = 2 + 8s, \quad y = 4 - 3s, \quad z = 5 + s, \quad s \in \mathbb{R}$$

For example, every point on line

$$L_1$$

is of the form

$$(1 + 4t, 5 - 4t, -1 + 5t)$$

. Which of the following is/are true?

Options :

☐ and are parallel

L_1

L_2

☐ and are not parallel

L_1

L_2

☐ and intersect at a point

L_1

L_2

☐ and do not intersect at any point

L_1

L_2

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 9 : 6406531951390

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following is/are true ?

Options :

- ☐ Every square matrix with real entries has at least one real eigenvalue.
- ☐ A set of two eigenvectors corresponding to the same eigenvalue of a square matrix must be linearly dependent.
- ☐ Given a square matrix of order n , the set of all eigenvectors for an eigenvalue λ , along with the zero vector, forms a subspace of $\mathbb{R}^n$.

n

λ

$\mathbb{R}^n$

- ☐ A square matrix is not invertible if and only if 0 is one of its eigenvalues.

0

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 10 : 6406531951389

View Solutions (0)Total Mark : 2.00 | Type : MSQ

Suppose you are building a credit card fraud detection system and have two approaches: (i) Manually writing rules such as "If transaction amount is above Rs. 50,000 and the location is overseas, mark it as fraud." (ii) Training a model using millions of past transactions labeled as fraud or not fraud. Which of the following options is/are correct?

Options :

- ☐ Approach (i) is not machine learning.
- ☐ Approach (ii) is machine learning.

- ☐ Both approaches are machine learning.
- ☐ Neither approach is machine learning.

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 11 : 6406531951391

View Solutions (0) Total Mark : 2.00 | Type : SA

Consider the function

$$f : \mathbb{R} \setminus \left\{ \frac{-1}{2} \right\} \rightarrow \mathbb{R}$$

given by

$$f(x) = \frac{1}{(1 + 2x)^5}$$

. If the best linear approximation to

$$f$$

at

$$x = 0$$

is given as

$$h(x) = a + bx$$

, find

$$a - b$$

. Your answer should be an integer.

Answer (Numeric):

Accepted Answer : 11

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 12 : 6406531951392

View Solutions (0)Total Mark : 3.00 | Type : SA

Find the directional derivative of the function

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}$$

given by

$$f(x, y, z) = x^2y - yz^3 + z$$

at the point

$$(1, -2, 0)$$

in the direction of the vector

$$\frac{1}{3} \begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$$

. Your answer should be an integer.

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 13 : 6406531951393

Total Mark : 0.00 | Type : COMPREHENSION

Let

$$P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be the projection matrix that projects vectors in

$$\mathbb{R}^2$$

onto the line

$$y = 3x$$

. Let

$$A = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

be the matrix that projects vectors in

$$\mathbb{R}^2$$

onto the nullspace of

$$P$$

. Based on the above data, answer the given subquestions.

Your score : 0

Broken Format?

Question 14 : 6406531951394

View Parent QNView Solutions (0)Total Mark : 3.00 | Type : SA

Find

$$10(a + b + c + d)$$

. Your answer should be an integer.

Answer (Numeric):

Accepted Answer : 16

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 15 : 6406531951395

View Parent QNView Solutions (0)Total Mark : 1.00 | Type : SA

Find

$$10(e + f + g + h)$$

. Your answer should be an integer.

Answer (Numeric):

Accepted Answer : 4

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 16 : 6406531951396

Total Mark : 0.00 | Type : COMPREHENSION

Consider the matrix

$$A$$

given below:

$$A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Your score : 0

Broken Format?

Question 17 : 6406531951397

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : SA

If

$$\lambda$$

is the smallest eigenvalue of

$$A$$

, enter the value of

$$\lambda$$

. Your answer should be an integer.

Answer (Numeric):

Accepted Answer : -3

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 18 : 6406531951398

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : SA

If

$$v = (a, -1)$$

is an eigenvector of

$$A$$

corresponding to

$$\lambda$$

, find

$$a$$

. Enter your answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 1.24 to 1.26

Your score : 0

Question 1 : 6406531527588

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 2 : 6406531527589

View Solutions (0) Total Mark : 1.00 | Type : MCQ

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

Regression

Classification

Clustering

Dimensionality Reduction

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 3 : 6406531527590

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2 .

Options :

$$y = x^2 + 1$$

$$y = x^2$$

$$y = x^2 - 1$$

$$y = 2x$$

Your score : 0

Discussions (1)Suggest Correction

Question 4 : 6406531527591

View Solutions (0)Total Mark : 2.00 | Type : MSQ

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$. Which of the following options is/are correct?

Options :

☐

Vectors v_1 and v_2 are parallel to each other.

☐

Vectors v_1 and v_2 are orthogonal to each other.

☐

Projection of v_1 onto v_2 is the zero vector.

☐

Projection of v_2 onto v_1 is the zero vector.

Your score : 0

Discussions (1)Suggest Correction

Question 5 : 6406531527592

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2, 4, 3), (5, 5, 6), (2, 2, 1), (4, 4, 6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1, x_2, x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u, u, u].$$

Find the reconstruction loss.

Hint : “The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input.”

Answer (Numeric):

Accepted Answer : 72

Your score : 0

Discussions (1)Suggest Correction

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Question 6 : 6406531527593

View Solutions (0)Total Mark : 3.00 | Type : SA

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Answer (Numeric):

Accepted Answer : 1024

Your score : 0

Discussions (1)Suggest Correction

Question 7 : 6406531527594

View Solutions (0)Total Mark : 2.00 | Type : SA

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A . What is the value of λ ?

Note that the term “column” here refers to a column vector, that is, each column of A is treated as a vector.

Answer (Numeric):

Accepted Answer : 0

Your score : 0

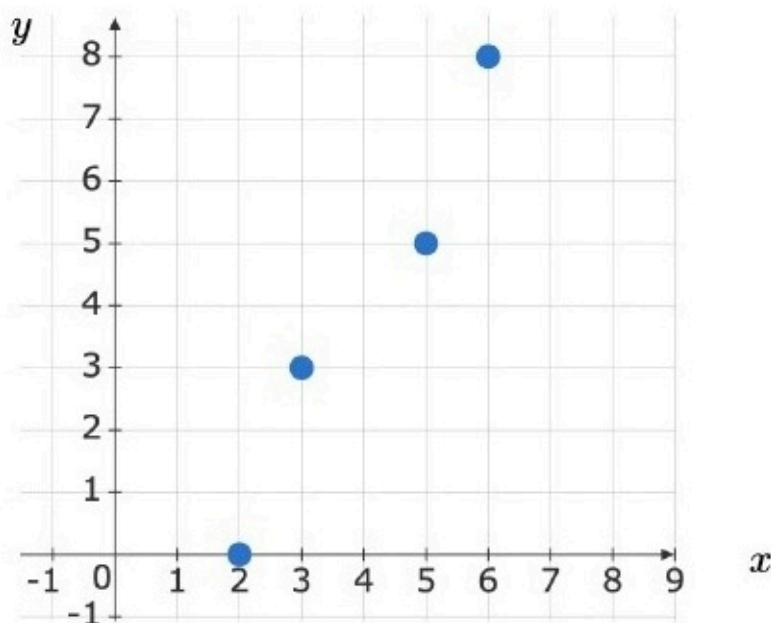
Discussions (1)Suggest Correction

Question 8 : 6406531527595

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Your score : 0

Broken Format?

Question 9 : 6406531527596

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MCQ

Which of these models has the smallest squared loss on this training dataset?

Options :

$$f(x) = 2x - 4$$

$$f(x) = 4$$

$$f(x) = 8$$

$$f(x) = 4 - 2x$$

Your score : 0

Discussions (1) [Suggest Correction](#) [Broken Format?](#)

Question 10 : 6406531527597

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : MCQ

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

No

Yes

Insufficient information to answer the question

Your score : 0

Discussions (2) [Suggest Correction](#)

Question 11 : 6406531527602

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Your score : 0

Broken Format?

Question 12 : 6406531527603

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MCQ

Which of the following sets forms a basis for the row space of A?

Options :

$\{(1, 0, 2), (0, 1, -1)\}$

$\{(1, 0, 2), (0, 1, -1), (0, 0, -1)\}$

$\{(1, 0, 2), (0, -1, 1), (0, -1, -4)\}$

$\{(0, 1, -1), (0, 0, -4)\}$

Your score : 0

Discussions (2)Suggest CorrectionBroken Format?

Question 13 : 6406531527604

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : MCQ

Which of the following linear relations
is valid among the rows (R_1, R_2, R_3)
of matrix A?

Options :

$$R_2 - R_1 - R_3 = 0$$

$$R_1 + R_2 - 2R_3 = 0$$

$$0.R_1 + 0.R_2 + 0.R_3 = 0$$

$$R_2 = 2R_1 + R_3$$

Your score : 0

Discussions (1)Suggest Correction

Question 14 : 6406531527605

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

where $0 < \theta < 180^\circ$.

Your score : 0

Broken Format?

Question 15 : 6406531527606

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : MSQ

Which of the following is/are true?

Options :

- ☐ A is an orthogonal matrix
- ☐ A is invertible
- ☐ A is symmetric

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 16 : 6406531527607

View Parent QNView Solutions (0)Total Mark : 3.00 | Type : MCQ

Which of the following is true?

Options :

A does not have real eigenvalues

A has two distinct real eigenvalues

A has two real eigenvalues, not necessarily distinct

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 17 : 6406531527608

Total Mark : 0.00 | Type : COMPREHENSION

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Your score : 0

Broken Format?

Question 18 : 6406531527609

View Parent QNView Solutions (0)Total Mark : 2.00 | Type : MCQ

If a square matrix A is diagonalizable, then A is invertible.

Options :

True

False

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 19 : 6406531527610

View Parent QNView Solutions (0)Total Mark : 1.00 | Type : MCQ

The eigenvalues of a real symmetric matrix are real.

Options :

True

False

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 20 : 6406531527611

View Parent QNView Solutions (0)Total Mark : 2.00 | Type : MCQ

A square matrix A of shape $n \times n$ is diagonalizable if and only if it has n distinct eigenvalues.

Options :

True

False

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 21 : 6406531527598

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Your score : 0

Broken Format?

Question 22 : 6406531527599

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : MCQ

Which of the following is
 $L_{(0,0)}[f](x)$?

Options :

$$1 + x_1 + x_2$$

$$x_1 + 2x_2$$

$$1 - x_1 - x_2$$

$$x_1 - x_2$$

Your score : 0

Discussions (1) [Suggest Correction](#)

Question 23 : 6406531527600

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : SA

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b ?

Answer (Numeric):

Accepted Answer : 1

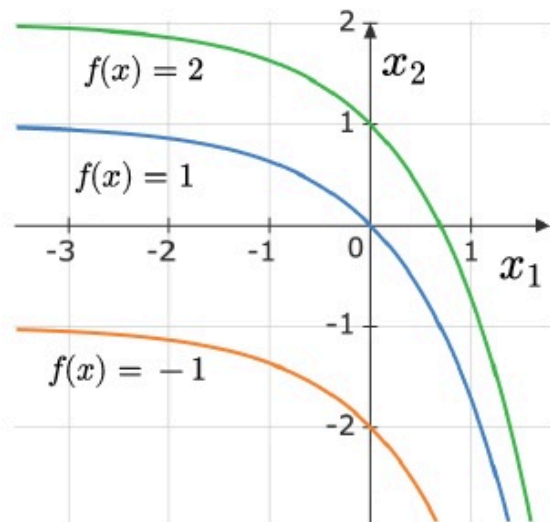
Your score : 0

Discussions (1) [Suggest Correction](#)

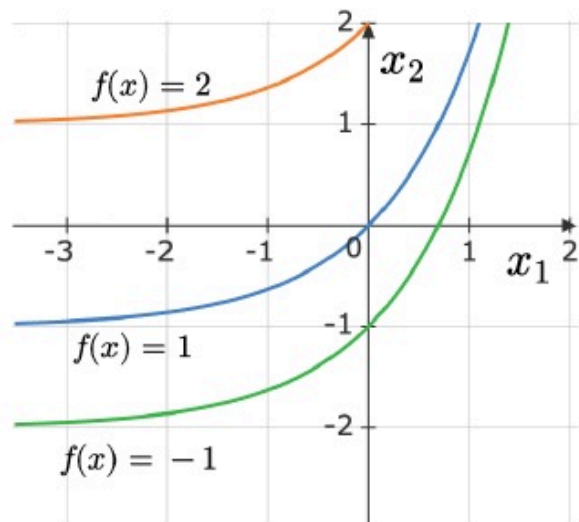
Question 24 : 6406531527601

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : MCQ

Which of the following represents the contours of f ?



(A)



(B)

Options :

(A)

(B)

Your score : 0

Question 1 : 6406531298181

Total Mark : 0.00 | Type : MCQ

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TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 6406531298182

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider a matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Which of the following options is/are true?

Options :

☐

Null space of A is span $\left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$

☐

Dimension of the column space of A is 1

☐

Left null space of A is span $\left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$



$$\text{Row space of } A \text{ is span } \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0.5 \end{bmatrix} \right\}$$

Your score : 0

Discussions (2)Suggest Correction

Question 3 : 6406531298188

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true regarding the projection matrix?

Options :

- ☐ It is a symmetric matrix.
- ☐ It is an orthogonal matrix.
- ☐ It is orthogonally diagonalizable.
- ☐ It is always a rank 1 matrix.

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 4 : 6406531298189

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider a matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Which of the following options is/are true?

Options :



1 is an eigenvalue of A.

☐

2 is an eigenvalue of A .

☐

$\begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix}$ is an eigenvector of A .

☐

$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector of A .

Your score : 0

Discussions (1)[Suggest Correction](#)

Question 5 : 6406531298190

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider a subspace $S = \{(x, y, z) \mid x + 2y + 2z = 0\}$ of the vector space $\mathbb{R}^3$.

Which of the following options is/are true?

Options :

☐

$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is a vector of S .

☐

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\} \text{ is an orthogonal basis of } S.$$

☐

The dimension of S is 1.

☐

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\} \text{ is a basis of } S.$$

Your score : 0

Discussions (3)Suggest Correction

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Question 6 : 6406531298194

View Solutions (0)Total Mark : 3.00 | Type : MSQ

The function

$$f(x) = \begin{cases} px^2 + qx + 2, & \text{if } x < 1 \\ rx + s, & \text{if } x \geq 1 \end{cases}$$

is differentiable at $x = 1$. If p, q, r , and s are constants, which of the following is/are must be true?

Options :

☐

$$r = 2p + q \text{ and } s = 2 - p$$

☐

$$p + q = r + s$$

☐

$$q = r \text{ and } s = 2$$

☐

$$p + q + 2 = r + s \text{ and } 2p + q = r$$

Your score : 0

Discussions (1)Suggest Correction

Question 7 : 6406531298195

View Solutions (0)Total Mark : 3.00 | Type : MSQ

The table below shows the original values and predicted values obtained by some machine learning algorithm for multiclass classification problems. Compute the squared error loss and 0-1 loss. Which of the following options regarding the loss Function is/are correct?

Original values	-1	0	1	2	3	1	0	3
Predicted values	1	1	1	2	1	0	0	3

Options :

- ☐ Square error loss $\leq$ 0-1 loss
- ☐ Square error loss $\geq$ 0-1 loss
- ☐ The 0-1 loss is 0.25
- ☐ The square error loss is 1.25

Your score : 0

Discussions (3)Suggest Correction

Question 8 : 6406531298196

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider a matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A = PDP^{-1}$.

Which of the following options is/are true?

Options :

☐

A is an orthogonally diagonalizable matrix.

☐

$$P = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

☐

$$P = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

☐

$$P^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

Your score : 0

Discussions (1) Suggest Correction

Question 9 : 6406531298183

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Let $\hat{y}$ be the regression line (the best fit line) for the dataset given below using the least squares method. Which of the following options is/are true?

x	0	2	3
y	-2	5	4

Options :

☐

$$\hat{y} = \frac{1}{14}(13x - 19)$$

☐

$$\hat{y} = \frac{1}{14}(31x - 91)$$

☐

If $x = 11$, then predicted value of y is 23.

☐

If $x = 7$, then predicted value of y is $\frac{99}{7}$.

Your score : 0

Discussions (1) [Suggest Correction](#)

Question 10 : 6406531298184

Total Mark : 0.00 | Type : COMPREHENSION

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Your score : 0

Question 11 : 6406531298185

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : MSQ

Which of the following options is/are true?

Options :

- ☐ $A^* = A$, where A^* is conjugate transpose of matrix A .
- ☐ A is a symmetric matrix.
- ☐ A is a unitary matrix.
- ☐ Columns of A are not orthogonal.

Your score : 0

Discussions (1) [Suggest Correction](#) [Broken Format?](#)

Question 12 : 6406531298186

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true ?

Options :

☐

1 is an eigenvalue of A .

☐

$1 + i\sqrt{2}$ is an eigenvalue of A .

☐

The rank of the matrix A is 2.

☐

Nullity of the matrix A is 1.

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 13 : 6406531298187

View Parent QN**View Solutions (0)** Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true ?

Options :

☐

Matrix A is diagonalizable.

☐

Eigenvectors of A are orthogonal.

☐

If A is diagonalizable and diagonal form of $A = P^{-1}DP$, then P is an orthogonal matrix.

☐

Entries of the eigenvector v
corresponding to the eigenvalue
 $1 - i\sqrt{2}$ of A are complex numbers.

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 14 : 6406531298191

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Which of the following is/are true?

Options :

☐

If $a_n = -1 + \frac{n}{n+1}$ then $a_n \rightarrow 0$ as $n \rightarrow \infty$.

☐

If $a_n = \sin(n)$ then $a_n \rightarrow 0$ as $n \rightarrow \infty$.

☐

If $a_n = n!$ then $a_n \rightarrow \infty$ as $n \rightarrow \infty$.

☐

$a_n = \sin(n)$ is not a convergent sequence.

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 15 : 6406531298192

View Solutions (0) Total Mark : 3.00 | Type : SA

Let $f(x, y) = \ln(x^2 + y^2)$. Use the linear approximation at the point $(1, 2)$ to estimate $f(1.02, 2.01)$. Write the answer correctly to 4 decimal places.

Answer (Numeric):

Accepted Answer : 1.6251 to 1.6258

Your score : 0

Discussions (1)Suggest Correction

Question 16 : 6406531298193

View Solutions (0)Total Mark : 2.00 | Type : MCQ

A hospital collects patient data including symptoms, vital signs, and lab test results. They want to find patterns to improve treatment plans. What type of task is clustering most suitable for?

Options :

Predicting the recovery time for a new patient.

Diagnosing diseases based on labeled medical records.

Grouping patients with similar symptoms and test results without prior labels.

Determining the most important variable for predicting disease.

Your score : 0

Question 1 : 6406531104744

Total Mark : 0.00 | Type : MCQ

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Options :

YES

NO

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 2 : 6406531104745

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the function $f(x, y) = \sin(xy)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ around the point $(\frac{\pi}{2}, 0)$. What is the value of $L(\frac{\pi}{2} + 0.1, 0.1)$?

hint: Use the approximation of $\pi = 3.14$

Options :

0.162

0.257

0.157

1.57

Your score : 0

Discussions (1)[Suggest Correction](#)

Question 3 : 6406531104746

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the matrix M given by:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Which of the following options is true?

Options :

Eigenvectors corresponding to the two different eigenvalues of M are orthogonal.

The matrix M is not diagonalizable.

The eigenvalues of M are 4 and 1

The rank of M is 1.

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 6406531104747

View Solutions (0)Total Mark : 3.00 | Type : MCQ

The directional derivative of a scalar function $f(x, y)$ at a point (x_0, y_0) in the direction of a unit vector u represents:

Options :

The rate of change of f in the direction of u at (x_0, y_0) .

The magnitude of the gradient vector ∇f at (x_0, y_0) .

The maximum value of the gradient at (x_0, y_0) .

The angle between the vector u and ∇f .

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 6406531104748

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Which of the following is the best second-order polynomial that fits the dataset by the least sum of squares method?

$$\min_{\theta_0, \theta_1, \theta_2} \sum_{i=1}^3 (y_i - (\theta_0 + \theta_1 x_i + \theta_2 x_i^2))^2$$

x	1	0	-1
y	-1	1	1

Options :

$$2 - \frac{1}{2}x + \frac{5}{2}x^2$$

$$2 - \frac{1}{2}x - \frac{5}{2}x^2$$

$$2 + \frac{1}{2}x - \frac{5}{2}x^2$$

$$2 + \frac{1}{2}x + \frac{5}{2}x^2$$

Your score : 0

Discussions (1)Suggest Correction

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Question 6 : 6406531104749

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true?

Options :

☐

$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is orthogonal to the vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

☐

For a matrix A , vectors of column space of A are perpendicular to the vectors of null space of A .

☐

Length of the vector, $a = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ is 3.

☐

$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ are orthogonal vectors.

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 7 : 6406531104750

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true when the vector $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ projected onto the vector $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$?

Options :

☐

The projection matrix is $\frac{1}{2} \begin{bmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

☐

The projection matrix is $\frac{1}{2} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

☐

Projected vector is $\frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

☐

Projected vector is $\frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

Your score : 0

Discussions (1)Suggest Correction

Question 8 : 6406531104752

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following statements is/are correct?

Options :

☐

$f(x) = \ln(x)$ is continuous and differentiable in the domain $[0, \infty)$.

☐

$f(x) = e^x$ is continuous and differentiable for all $x \in \mathbb{R}$.

☐

$f(x) = \sqrt{x}$ is continuous but not differentiable at $x = 0$.

☐

$f(x) = \frac{1}{x}$ is continuous and differentiable for all $x \neq 0$.

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 9 : 6406531104751

View Solutions (0) Total Mark : 2.00 | Type : MSQ

Consider a matrix

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 15 \end{bmatrix}.$$

Which of the following options is/are true?

Options :

☐

Null space of A is $\text{span} \left\{ \begin{bmatrix} -5 \\ 2 \end{bmatrix} \right\}$

☐

Column space of A is span $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$

☐

Left null space of A is span $\left\{ \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$

☐

Row space of A is span $\left\{ \begin{bmatrix} 2 \\ -5 \end{bmatrix} \right\}$

Your score : 0

Discussions (1)Suggest Correction

Question 10 : 6406531104753

View Solutions (0)Total Mark : 2.00 | Type : SA

Consider a matrix $A_{20 \times 25}$. If the nullity of the matrix is 10, then find the rank of the matrix.

Answer (Numeric):

Accepted Answer : 15

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 6406531104754

View Solutions (0)Total Mark : 3.00 | Type : SA

If $f(x, y, z) = \sin(xy) + e^{yz} + xz^2$, what is the length of the gradient at the point, $(1, 0, 2)$? Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 6.37 to 6.43

Your score : 0

Discussions (1)Suggest Correction**Question 12 : 6406531104755**

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

A bank wants to predict whether a customer will default on a loan using a classification model. The bank collects data from past customers, which includes the following features:

- x_1 : Credit score (scaled between 0 and 1)
- x_2 : Monthly income (in \$1000s)
- x_3 : Number of late payments in the past year

The target variable y represents whether the customer defaulted (1 for default, 0 for no default). The data is provided in the table below:

x	y
[0.9, 5, 1]	0
[0.4, 3, 5]	1
[0.8, 4, 2]	0
[0.2, 2, 7]	1
[0.7, 6, 0]	0

The bank uses the following linear combination to compute z :

$$z = 0.5x_1 + 0.3x_2 - 0.2x_3$$

The step function $u(z)$ is used to classify the outcome:

$$u(z) = \begin{cases} 1 & \text{if } z \geq 1.5 \\ 0 & \text{otherwise} \end{cases}$$

Your score : 0

Broken Format?

Question 13 : 6406531104756

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MCQ

For the input $x = [0.9, 5, 1]$, what will be the predicted output $u(z)$?

Options :

0

1

1.75

Cannot be determined

Your score : 0

Discussions (1)Suggest CorrectionBroken Format?

Question 14 : 6406531104757

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MCQ

If the threshold in the step function is changed to $z \geq 1.0$, what will happen to the misclassification rate?

Options :

Increase

Decrease

Remain same

Cannot be determined

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 15 : 6406531104758

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider a 3×3 matrix B given by

$$B = (-1) \begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Your score : 0

Broken Format?

Question 16 : 6406531104759

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MCQ

If B is diagonalizable, what is the invertible matrix P such that $B = PDP^{-1}$?

Options :

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

B is not diagonalizable.

Your score : 0

Discussions (0) **Suggest Correction** **Broken Format?**

Question 17 : 6406531104760

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MCQ

If B is diagonalizable, which of the following is a diagonal matrix D such that $B = PDP^{-1}$? Use the matrix P obtained in the previous question.

Options :

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

B is not diagonalizable.

Your score : 0

Question 1 : 640653995423

Total Mark : 0.00 | Type : MCQ

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Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653995424

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following 3×3 matrices have rank 2 with column space equal to row space?

Options :

☐

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

☐

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 3 : 640653995428

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

- ☐ B has eigenvalues -4, 8, 70.
- ☐ A has eigenvalues 1, 16, 81.
- ☐ B is an upper triangular matrix.
- ☐ B is a lower triangular matrix.

Your score : 0

Discussions (1)[Suggest Correction](#)

Question 4 : 640653995425

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Consider a matrix $A = \begin{bmatrix} 6 & 3 \\ -2 & 1 \end{bmatrix}$ and a set $S = \{v_1 = (1, -1), v_2 = (1, 1)\}$.
Which of the following options is/are correct?

Options :

☐

S is the set of eigenvectors of A .

☐

Only v_1 is an eigenvector of A .

☐

$$A^{50} = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} -2 & -3 \\ 1 & 1 \end{pmatrix}$$

☐

$$A^{50} = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$$

Your score : 0

Discussions (1)Suggest Correction

Question 5 : 640653995426

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Consider two vectors $a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Let P be the projection

matrix that projects a vector onto the line through the vector a . Which of the following options is/are true?

Options :

☐

P is symmetric matrix.

☐

Projection of b onto a is $\frac{1}{4}(-a)$.

☐

P is rank one matrix.

☐

Plane $x = y$ is orthogonal to a .

Your score : 0

Discussions (0)[Suggest Correction](#)

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Question 6 : 640653995427

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Which of the following options is/are correct for the following dataset?

x	1	1	-1
y	1	1	1

Options :

☐

The best-fit line for the dataset given using the least squares method is, $\hat{y} = 1$.

☐

The best-fit line for the dataset given using the least squares method is, $\hat{y} = -1$.

☐

The best-fit line for the dataset given using the least squares method is, $\hat{y} = x - 1$.

☐

The best-fit line for the dataset given using the least squares method is, $\hat{y} = x + 1$.

☐

Error vector (the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$.

☐

Error vector(the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653995433

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Determine which of the following limits yields a finite value.

Options :

☐

$$\lim_{x \rightarrow 1} \frac{x^{\frac{1}{3}} - 1}{x^{\frac{1}{2}} - 1}$$

☐

$$\lim_{x \rightarrow 1} \frac{1 - x^{\frac{1}{2}}}{1 - x}$$

☐

$$\lim_{x \rightarrow \infty} \frac{x^2 - 7x}{x + 1}$$

☐

$$\lim_{x \rightarrow a} \frac{x^2 - a^2}{x^4 - a^4}, \text{ where } a > 0$$

☐

None of these

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 8 : 640653995429

View Solutions (0) Total Mark : 3.00 | Type : MCQ

$$\text{Let } A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 6 \end{bmatrix}. \text{ Is } A \text{ diagonalizable?}$$

Options :

Yes

No

Insufficient information

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653995430

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following input data points:

X	Y
$[0, 1, 1]$	-1
$[0, 1, 0]$	-1
$[1, 1, 3]$	+1
$[1, 0, 5]$	+1
$[0, 0, 6]$	+1
$[1, 0, 4]$	+1

Which among the following models has the least misclassification error?

Hint: Consider $Sign(0) = 1$

Options :

$$Sign(x_1 + x_2 + x_3 - 5)$$

$$Sign(x_1 + x_2 - 2)$$

$$Sign(x_1 + x_2 + x_3)$$

$$Sign(x_3 - x_2 - x_1)$$

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653995432

View Solutions (0)Total Mark : 3.00 | Type : MCQ

For what value of b is

$$f(x) = \begin{cases} \frac{x-b}{b+1}, & \text{if } x < 0 \\ x^2 + b, & \text{if } x \geq 0 \end{cases}$$

continuous at every x ?

Options :

3

2

•

◦ 2

1

Your score : 0

Discussions (1)Suggest Correction

Question 11 : 640653995435

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the function, $f(x, y) = x^2 + 3y^2$, in which direction does the maximum directional derivative occur at the point $(1, 2)$?

Options :

$$\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}} \right)$$

$$\left(\frac{1}{\sqrt{37}}, \frac{6}{\sqrt{37}} \right)$$

$$\left(\frac{2}{\sqrt{5}}, \frac{24}{\sqrt{5}} \right)$$

$$\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$$

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 640653995431

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider the function, $f(x, y) = e^x \cos(y)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ at the point $(0, 0)$, then find the value $L(0.5, 0)$. Enter the answer correct to one decimal place.

Answer (Numeric):

Accepted Answer : 1.5

Your score : 0

Discussions (0)Suggest Correction

Question 13 : 640653995434

View Solutions (0)Total Mark : 3.00 | Type : SA

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function, $f(x) = 1 + e^{-x}$. What is the value of the derivative of f at x where $f(x) = 1.4$? Enter the answer correct to one decimal place.

Answer (Numeric):

Accepted Answer : 0.4

Your score : 0

Question 1 : 640653827349

Total Mark : 0.00 | Type : MCQ

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Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653827350

Total Mark : 0.00 | Type : COMPREHENSION

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A . Based on this information, answer the given sub-questions.

Your score : 0

Question 3 : 640653827351

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : SA

Find the value of $\alpha + 2\beta$.

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (1) [Suggest Correction](#) [Broken Format?](#)

Question 4 : 640653827352

View Parent QNView Solutions (0) Total Mark : 4.00 | Type : MSQ

Which of the following options is/are true ?

Options :

☐

Eigenvectors, v_1 and v_2 are linearly independent.

☐

Eigenvectors, v_1 and v_2 are orthogonal.

☐

A third eigenvector of A is $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

☐

$$a + b + c = 9.$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 5 : 640653827353

Total Mark : 0.00 | Type : COMPREHENSION

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Your score : 0

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Question 6 : 640653827354

[View Parent QN](#)[View Solutions \(0\)](#)Total Mark : 4.00 | Type : MSQ

Which of the following options is/are true?

Options :

☐ A is a symmetric matrix.

☐ A is a diagonal matrix.

- ☐ A is orthogonally diagonalizable.
- ☐ Eigenvalues of A are -1, -1 and 2.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 7 : 640653827355

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MSQ

Let v_1 and v_2 are the eigenvectors of A corresponding to distinct eigen-values.

We construct a new matrix B using these eigenvectors as its columns, such that

$B = [v_1, v_2]$. Based on this information, which of the following options are correct?

Options :

☐

Null space of the matrix B is zero space.

☐

Column space of the matrix B is $\mathbb{R}^3$.

☐

Column space of the matrix B represents a plane in $\mathbb{R}^3$.

☐

Null space of the matrix B represents
a line in $\mathbb{R}^3$.

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653827356

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Let P be the projection matrix for a vector $a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Which of the following options is/are true?

Options :

☐

P is orthogonally diagonalizable.

☐

$$P^4 = P.$$

☐

Projection of $b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ on the vector a is $-\frac{1}{5}a$.

☐

$$\text{Rank}(P) = 2$$

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653827357

View Solutions (0) Total Mark : 4.00 | Type : MCQ

Find the best-fit line for the dataset given below using the least squares method.

x	0	2	3	5
y	1	2	4	3

Options :

$$\hat{y} = 1.34 + 0.46x$$

$$\hat{y} = 0.46 + 1.34x$$

$$\hat{y} = 1.34 - 0.46x$$

$$\hat{y} = 2.16 + 0.74x$$

Your score : 0

Discussions (0) **Suggest Correction**

Question 10 : 640653827358

View Solutions (0) Total Mark : 4.00 | Type : MCQ

The function

$$f(x) = \begin{cases} ax^2 + bx + 1, & \text{if } x < 0 \\ cx + d, & \text{if } x \geq 0 \end{cases}$$

is differentiable at $(x = 0)$. If a , b , c , and d are constants, which of the following must be true?

Options :

$$a = c \text{ and } b = 1$$

$$b = c \text{ and } d = 1$$

$$a = c \text{ and } d = 1$$

$$b = d = 1$$

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 640653827359

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Which of the following statements is true for a function $f(x, y)$ that is differentiable at a point (a, b) ?

Options :

The directional derivative exists in every direction.

The directional derivative is the same in every direction.

The directional derivative can only be calculated along the axes.

The directional derivative does not exist.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 12 : 640653827360

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Which of the following functions cannot be approximated linearly around $x = 0$?

Options :

$$f(x) = \sin(x)$$

$$f(x) = \cos(x)$$

$$f(x) = \ln(x)$$

$$f(x) = x^2$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 13 : 640653827361

View Solutions (0)Total Mark : 3.00 | Type : SA

If $f(x, y) = x^2y + 3xy^2$, what is the length of the gradient at the point (1,2)?
Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 20.58 to 20.64

Your score : 0

Discussions (1)[Suggest Correction](#)

Question 14 : 640653827362

View Solutions (0)Total Mark : 3.00 | Type : SA

In a sports analytics company, the team is interested in predicting whether a basketball player will score above 20 points in a game using classification models. They collected data on various factors $x = [x_1, x_2, x_3]$ from past games of different players. The factors include minutes played, shots attempted, and free throw percentage. The data and corresponding labels are shown in the table below: Compute the misclassification

x	y
[1, 5, 0.8]	1
[0, 3, 0.9]	0
[1, 2, 0.7]	0
[1, 8, 0.6]	1
[0, 7, 0.5]	0

rate if they use the following step function:

$$u(z) = \begin{cases} 1, & \text{if } z \geq 3 \\ 0, & \text{otherwise} \end{cases}$$

and the linear combination z is given by: $0.6x_1 + 0.3x_2 + 0.1x_3$. Enter the answer correct to one decimal place.

Note: Here, x_1 represents whether the player is a starter (1) or not (0), x_2 is the number of shots attempted, and x_3 is the free throw percentage. The label y indicates whether the player scored more than 20 points (1) or not (0). The misclassification rate is the number of labels misclassified.e.g. if 3 out of 5 players are mislabeled then the misclassification rate is $3/5$.

Answer (Numeric):

Accepted Answer : 0.2

Your score : 0

Discussions (1)Suggest Correction

Question 15 : 640653827363

View Solutions (0)Total Mark : 2.00 | Type : MCQ

Consider a dataset from an e-commerce company such as Flipkart that has details of customer behavior, which of the following problems can be best approached using clustering?

Options :

Predicting the exact number of sales for each customer segment.

Classifying customers into segments based on labeled customer data.

Identifying inherent groupings in customer data without predefined labels.

Reducing the number of features in the customer dataset.

Your score : 0

Question 1 : 640653737983

Total Mark : 0.00 | Type : MCQ

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Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653737984

Total Mark : 0.00 | Type : COMPREHENSION

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3×3 matrix. Based on this information, answer the given subquestions:

Your score : 0

Question 3 : 640653737985

View Parent QNView Solutions (0) Total Mark : 4.00 | Type : MCQ

Find the row space of matrix A.

Options :

$$\text{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix} \right\}$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix} \right\}$$

Your score : 0

Discussions (1) [Suggest Correction](#) [Broken Format?](#)

Question 4 : 640653737986

View Parent QNView Solutions (0) Total Mark : 3.00 | Type : MCQ

Find the column space of matrix A.

Options :

$$\text{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix} \right\}$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}, \begin{pmatrix} 3 \\ -3 \\ 6 \end{pmatrix} \right\}$$

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix} \right\}$$

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 5 : 640653737989

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

$$\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & -2 \\ 0 & 0 \end{bmatrix}$$

Your score : 0

Discussions (0)Suggest Correction

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Question 6 : 640653737995

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Which of the following is the best approximation of $\sqrt[4]{16.1}$? (Use linear approximation around 16).

Options :

2.1

2.003125

2.031

1.9968

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653737990

View Solutions (0)Total Mark : 4.00 | Type : MCQ

Find the best fit line for the dataset given below using least squares method.

x	0	2	3	6	8
y	-2	5	3	1	-3

Find the best fit line for the dataset given below using least squares method.

Options :

$$\hat{y} = 0.6x - 3.2$$

$$\hat{y} = -0.6x + 3.2$$

$$\hat{y} = -x + 2.8$$

$$\hat{y} = -0.4x + 2.8$$

Your score : 0

Discussions (2)[Suggest Correction](#)[Broken Format?](#)

Question 8 : 640653737991

View Solutions (0) Total Mark : 4.00 | Type : MCQ

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a multivariable scalar-valued continuous function, such that

$$\frac{\partial f}{\partial x} = 2x + 1 \text{ and } \frac{\partial f}{\partial y} = 2y + 1.$$

Suppose $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 5$.

Find out the directional derivative of f at the point $(0,0)$ in the direction of the vector $(1,1)$.

Options :

$$\sqrt{2}$$

$$2\sqrt{2}$$

$$-2\sqrt{2}$$

$$-\sqrt{2}$$

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 9 : 640653737993

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following is/are supervised learning task(s)?

Options :

- ☐ Predicting future sales of a product based on the past sales data.
- ☐ Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.
- ☐ Recognizing and interpreting handwritten characters or words, often used in digit recognition or reading handwritten text.
- ☐ Categorizing text documents into predefined classes, such as spam or non-spam emails.

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 10 : 640653737987

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

- ☐ Eigenvalues of A are 1, 2, and 3.
- ☐ Eigenvalues of A are 1, 2, and 2.
- ☐ Eigenvalues of B are 2, 20, and 78.
- ☐ Matrix B is diagonalizable.

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 11 : 640653737988

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Let A and B be symmetric matrices such that $AB = BA$. Then which of the following options is/are true?

Options :

- ☐ AB is a symmetric matrix.
- ☐ AB is orthogonally diagonalizable.
- ☐ BA has real eigenvalues.
- ☐ AB is not a symmetric matrix.

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 12 : 640653737994

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Consider a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined as $f(x) = |x(x - 2)|$

Which of the following options is/are true?

Options :

☐

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x).$$

☐

f is differentiable at $x = 0$.

☐

$$f'(3) = 4.$$

☐

f is continuous at $x = 0$.

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 13 : 640653737992

View Solutions (0)Total Mark : 4.00 | Type : SA

Consider an encoder-decoder pair used for dimensionality reduction, where the encoder function is denoted as $f(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_1 + x_3)$ and the decoder function is denoted as $g(u) = (u_1, u_2, u_1 + u_2)$

Compute the reconstruction error $\left(R(f, g) = \frac{1}{n} \sum_{i=1}^n \|X_i - g(f(X_i))\|^2 \right)$ for the following dataset:

$$X_1 = [1, 2, 3], \quad X_2 = [0, 2, 2], \quad X_3 = [1, 0, -1]$$

Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 3.12 to 3.20

Your score : 0

Question 1 : 640653669180

Total Mark : 0.00 | Type : MCQ

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Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653669181

View Solutions (0)Total Mark : 2.00 | Type : MSQ

Which of the following is/are supervised learning task(s)?

Options :

- ☐ Predicting future weather conditions based on the past weather observations.
- ☐ Reducing the dimensionality of images while retaining essential features,enabling efficient storage and transmission.
- ☐ Identifying objects in images with multiple categories, such as cats, dogs, cars,and trees.
- ☐ Categorizing news articles into topics like politics, sports, entertainment, andtechnology.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 3 : 640653669182

View Solutions (0)Total Mark : 2.00 | Type : MCQ

Consider the following input data points:

X	Y
$[0, 1]$	$+1$
$[0, 2]$	$+1$
$[1, 3]$	$+1$
$[3, 4]$	-1
$[4, 5]$	-1
$[7, 6]$	-1

Consider the following input data points: Which among the following models has the least misclassification error?

Options :

Sign($x_1 - 2$)

Sign($x_2 + 2$)

Sign($x_1 + x_2$)

Sign($x_2 - x_1$)

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 4 : 640653669188

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Consider a vector $v = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Does v belong to the columnspace of $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$?

Options :

Yes

No

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 5 : 640653669194

View Solutions (0) Total Mark : 2.00 | Type : MCQ

Let $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

$$\begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}$$

Your score : 0

Discussions (0)Suggest Correction

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Question 6 : 640653669183

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider a regression model $h(x_1, x_2) = \frac{x_1 + x_2}{2}$. For the below input data points, find the mean squared loss of h . Enter the answer correct to two decimal places.

X	Y
$[1, 2]$	2
$[2, 3]$	3
$[3, 4]$	4
$[4, 5]$	5

Answer (Numeric):

Accepted Answer : 0.25

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653669185

View Solutions (0)Total Mark : 3.00 | Type : SA

The curve $y = ax^2 + bx + c$ passes through the points $(2, -1)$ and $(0, 0)$, and is tangent to the line $y = -x$ at the origin. Find the absolute value of the product ab . Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 0.25

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653669186

View Solutions (0)Total Mark : 3.00 | Type : SA

At the point $P_0 = (1, 2)$, the function $f(x, y)$ has a derivative of $2\sqrt{2}$ in the direction $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and a derivative of 5 in the direction $(0, 1)$. Find the derivative of $f(x, y)$ at point P_0 in the direction $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$. Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 4.88 to 4.95

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653669195

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider a 2×2 matrix A whose eigenvectors are $v_1 = [1, 1]^T$ and $v_2 = [1, 2]^T$ with eigenvalues 2 and 3, respectively. Find the sum of square of all the entries of A .

Answer (Numeric):

Accepted Answer : 22

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653669184

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Let $f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases}$. Select which among the following are true?

Options :

- ☐ f is discontinuous at $x = 0$.
- ☐ f is differentiable.
- ☐ f is not differentiable at $x = 0$.
- ☐ $f'(0) = 0$

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 640653669187

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Find the second order approximation for $f(x, y) = e^{2y-x}$ in the neighbourhood of $(0, 0)$.

Options :

$$1 - x + 2y - 2xy + y^2 + \frac{x^2}{2}$$

$$1 - 2xy + 2y^2 + \frac{x^2}{2}$$

$$2y^2 + \frac{x^2}{2}$$

$$1 - x + 2y - 2xy + 2y^2 + \frac{x^2}{2}$$

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 640653669193

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Find the best fit line for the dataset given below using least squares method.

x	1	3	4	5	7
y	-3	7	1	0	4

Options :

$$\hat{y} = 0.7x - 1$$

$$\hat{y} = x + 0.5$$

$$\hat{y} = x + 0.2$$

$$\hat{y} = 0.2x + 1$$

Your score : 0

Discussions (0)Suggest Correction

Question 13 : 640653669196

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Let A be a $n \times n$ real matrix. Then, which among the following statements are true?

Options :

If the eigenvalues of A are $\lambda_1, \dots, \lambda_n$, then A is similar to a diagonal matrix with $\lambda_1, \dots, \lambda_n$ along the diagonal.

If $\text{rank}(A) = r$, then A has r non-zero eigenvalues.

A may have complex eigenvalues.

If A has a repeated eigenvalue, then A is not diagonalizable.

Your score : 0

Discussions (0)Suggest Correction

Question 14 : 640653669189

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Select the correct options from the following.

Options :

☐

There exists a matrix whose row space contains $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and whose null space contains $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

☐

There exists a matrix with row space $= \mathbb{R}^3$ and column space $= \mathbb{R}^4$.

☐

There exists a matrix whose column space contains $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and row space contains $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$



There exists a matrix whose nullspace consists of all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 15 : 640653669190

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \in \mathbb{R}^3$, where

$$U = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}, \quad b = \begin{bmatrix} 2 \\ 0 \\ -4 \end{bmatrix}$$

Your score : 0

Broken Format?

Question 16 : 640653669191

View Parent QNView Solutions (0)Total Mark : 3.00 | Type : MCQ

Find the projection of b onto U.

Options :

$$\begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ -1 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 17 : 640653669192

View Parent QN**View Solutions (0)** Total Mark : 1.00 | Type : SA

Find the length of the error vector $e = b - p$, where p is the projection of b onto U .
Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 3.70 to 3.78

Your score : 0

Question 1 : 640653577678

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : MACHINE LEARNING** FOUNDATIONS (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT

SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653577679

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following statements are correct?

Options :

☐

$$I(444\%3 = 0 \text{ and } \pi \in \mathbb{R}) = 1.$$

☐

Mean Squared Error (MSE) is a suitable loss function for classification problems.

☐

The loss obtained after applying encoder and decoder functions to a given data can be negative.

☐

Gaussian mixture model is a family of distributions for modeling multimodal data.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 3 : 640653577691

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following statements is/are true?

Options :

- ☐ Row space of a matrix is orthogonal to the null space of the matrix.
- ☐ Row space of a matrix is orthogonal to the null space of transpose of thematrix.
- ☐ Dimension of row space of a matrix is always equal to the dimension of column space of the matrix.
- ☐ Dimension of row space of a matrix need not be equal to the dimension of column space of the matrix.

Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 4 : 640653577693

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following matrices have eigenvectors as $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$?

Options :

☐

$$\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

☐

$$\begin{bmatrix} 6 & 4 & 8 \\ 4 & 0 & 4 \\ 8 & 4 & 6 \end{bmatrix}$$

☐

$$\begin{bmatrix} 2 & 1 & 4 \\ 4 & 0 & 4 \\ 8 & 4 & 6 \end{bmatrix}$$

☐

$$\begin{bmatrix} 3 & 1 & 0 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 5 : 640653577694

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Let the matrix $A \in \mathbb{R}^{d \times d}$ be diagonalizable, then which of the following statements is/are true for A ?

Options :

☐

A must be symmetric.

☐

We can find a basis of eigenvectors for $\mathbb{R}^d$.

☐

We can find an orthonormal basis of eigenvectors for $\mathbb{R}^d$.



A can have some repeated eigenvalues but must have distinct independent eigenvectors.

Your score : 0

Discussions (0)Suggest Correction

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Question 6 : 640653577686

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Let $A, B \in \mathbb{R}^{n \times n}$ be two matrices, then which among the following statements are correct?

Options :



$$x \in \text{Nullspace}(A) \implies x \in \text{Nullspace}(BA).$$



$$x \in \text{Columnspace}(A) \implies x \in \text{Columnspace}(B), \text{ where } A = BC, C \in \mathbb{R}^{n \times n}.$$



$$x \in \text{Nullspace}(AB) \implies Bx \in \text{Nullspace}(A).$$



$$x \in \text{Nullspace}(AB) \implies x \in \text{Nullspace}(B).$$

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653577692

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Consider the solution of the linear regression problem ($Ax = b$) using least squares method, where $A = \begin{bmatrix} 1 & 5 & 5 \\ 2 & 3 & 5 \\ 0 & 7 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 7 \\ 8 \\ 4 \end{bmatrix}$.

Based on this data, which of the following statements is/are true?

Options :

☐

The error value is zero as we can get an x such that Ax is exactly equal to b .

☐

The error value is non-zero as we cannot get an x such that Ax is exactly equal to b .

☐

Least squares solution for x is $\begin{bmatrix} 2 \\ 0.5 \\ 0.5 \end{bmatrix}$.

☐

Least squares solution for x is $\begin{bmatrix} 1 \\ 0.6 \\ 0.4 \end{bmatrix}$.

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653577684

View Solutions (0)Total Mark : 4.00 | Type : MCQ

Find the linear approximation of the function $f(x, y) = ye^{xy}$ in the neighbourhood of $(0, 1)$.

Options :

$$x + y$$

$$x + 2y$$

$$2x + y$$

$$2x + 3y$$

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653577685

View Solutions (0)Total Mark : 4.00 | Type : MCQ

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, given by $f(x, y) = x^2 + y^2$. Consider the following algorithm:

$$X_{n+1} = X_n - h \nabla f(X_n), \quad X_i \in \mathbb{R}^2, i : 1, 2, \dots \text{ and } h > 0,$$

where h is the step size and $\nabla f(X_n)$ is the gradient of f evaluated at X_n . Fix $h = 0.25$ and $X_0 = (1, 3)$, which among the following points does the algorithm converge to?

Options :

$$(1, 3)$$

$$(1/2, 1/6)$$

$$(0, 0)$$

The algorithm does not converge

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653577680

View Solutions (0)Total Mark : 4.00 | Type : SA

Consider the following dataset with 5 data points:

X	y
(2, 3)	5
(-1, 1)	2
(4, 2)	7
(0, -2)	1
(-3, 5)	4

We want to fit a linear regression model of the form $f(X) = w^T x$ to this dataset, where $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Compute the value of the loss function L for this dataset which is defined as $L = \max_i |f(x^i) - y^i|$.

Answer (Numeric):

Accepted Answer : 5

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 640653577681

View Solutions (0)Total Mark : 4.00 | Type : SA

Let $f(x_1, x_2, x_3) = \frac{x_1 + x_2 + x_3}{2}$ be used as an encoder function and $g(u) = [u, u, u]$ be used as a decoder function for dimensionality reduction of the dataset $X_1 = [1, 2, 3]$, $X_2 = [2, 4, 6]$, $X_3 = [3, 6, 9]$, $X_4 = [4, 8, 12]$. Calculate the reconstruction error $\left(R(f, g) = \frac{1}{n} \sum_{i=1}^n \|x_i - g(f(x_i))\|^2 \right)$ for this encoder decoder pair. Enter the answer correct to one decimal place.

Answer (Numeric):

Accepted Answer : 37 to 38

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 640653577690

View Solutions (0)Total Mark : 4.00 | Type : SA

Let P1 be the projection matrix onto the line spanned by the vector $a = [1, 2, 3]^T$.
Let P2 be the projection matrix that projects vectors onto the plane perpendicular to span of a . Find the determinant of $P1 + P2$.

Answer (Numeric):

Accepted Answer : 1

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 13 : 640653577682

View Solutions (0)Total Mark : 3.00 | Type : SA

Find the directional derivative of $f(x, y, z) = x(x^2 - y^2) - z$ at a point $D(1, -1, 0)$ in the direction of vector $\hat{h} = \frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k$. Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : -1.16 to -1.12

Your score : 0

Discussions (0)Suggest Correction

Question 14 : 640653577683

View Solutions (0)Total Mark : 2.00 | Type : SA

Let the function $f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$

be continuous for all $x \in \mathbb{R}$. Then, find the value of a .

Enter the answer correct to one decimal place.

Answer (Numeric):

Accepted Answer : 3.5

Your score : 0

Discussions (0)Suggest Correction

Question 15 : 640653577687

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$.

Your score : 0

Broken Format?

Question 16 : 640653577688

View Parent QNView Solutions (0) Total Mark : 2.00 | Type : MCQ

Is the system $Ax = b$ consistent?

Options :

Yes

No

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 17 : 640653577689

View Parent QNView Solutions (0)Total Mark : 3.00 | Type : SA

Using the least square approximation, find the length of the projection vector p of b onto the column space of A . Enter the answer correct to two decimal places.

Answer (Numeric):

Accepted Answer : 3.12 to 3.16

Your score : 0

Question 1 : 640653388551

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL: MACHINE LEARNING FOUNDATIONS"

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Options :

Yes

No

Your score : 0

Discussions (0)Suggest Correction

Question 2 : 640653388552

View Solutions (0)Total Mark : 4.00 | Type : SA

Note: Enter the answer correct to two decimal places.

The ML team in a movie production company wanted to predict whether a movie will earn 100 cr or not by using some classification models. Therefore, the team collected presence/absence of various factors $\mathbf{x} = [x_1, x_2, x_3]$ from the movies in the past. The data and the corresponding labels are shown in the Table below.

$\mathbf{x}$	y
$[0,1,1]$	0
$[1,0,1]$	1
$[1,0,0]$	1
$[1,1,1]$	1
$[1,1,0]$	1

Compute the loss of the model if they use

$$u(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

and $z = 0.5x_1 - x_2 + 0.4x_3$

Answer (Numeric):

Accepted Answer : 0.38 to 0.42

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 3 : 640653388558

View Solutions (0)Total Mark : 4.00 | Type : SA

Find the linear approximation (L) of $f(x, y) = 3x^3 + 4y^2 + 10$ around (1, 1) and use it to compute $L(1.2, 0.8)$. What will be the value of $f(1.2, 0.8) - L(1.2, 0.8)$? Enter the answer up to two decimals accuracy.

Answer (Numeric):

Accepted Answer : 0.50 to 0.58

Your score : 0

Discussions (1)[Suggest Correction](#)

Question 4 : 640653388565

View Solutions (0)Total Mark : 4.00 | Type : SA

If the second degree polynomial that fits the below is $ax^2 + bx + c$, then what is the value of $a + b + c$?

x	y
1	3
2	4
3	8

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 640653388553

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Identify which of the following requires use of classification technique.

Options :

- ☐ Credit card fraud detection.
- ☐ Sentiment analysis
- ☐ Predicting number of emails a user will receive in a single day.
- ☐ Predicting whether the patient is having cancer or not.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

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Question 6 : 640653388560

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Two vectors v_1 and v_2 are such that $v_1 = -2v_2$. If $v_1 = (1, 2, 3)^T$, and we have another vector $v_3 = (1, -5, 3)^T$, then which of the following statements is/are true?

Options :

☐

Vectors v_1 and v_2 are parallel to each other.

☐

Vectors v_1 and v_2 are perpendicular to each other.

☐

Vectors v_2 and v_3 are not perpendicular to each other.

☐

Vectors v_2 and v_3 are perpendicular to each other.

☐

Vectors v_1 and v_3 are perpendicular to each other.

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653388563

View Solutions (0)Total Mark : 4.00 | Type : MSQ

Which of the following is/are not the eigen vectors of the matrix $\begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix}$?

Options :

☐

$$(1, -1)^T$$

☐

$$(1, 1)^T$$

☐

$$(1, 1.5)^T$$

☐

$$(2, 3)^T$$

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653388554

View Solutions (0)Total Mark : 3.00 | Type : SA

For the data sets $(x^i, y^i) = [(1, 1), (2, 2), (3, 4), (4, 5), (5, 5)]$, $i = 1$ to 5, Consider the regression model $f(x) = x + 1$. What is the mean squared loss of $f(x)$. (Enter answer correct to one decimal place)

Answer (Numeric):

Accepted Answer : 0.6

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653388556

View Solutions (0)Total Mark : 3.00 | Type : SA

What will be the directional derivative of $f(x, y, z) = xyz$ at point $(-1, 1, 2)$ along the direction given by the unit vector $(\frac{1}{3}, \frac{-2}{3}, \frac{2}{3})$ Enter the answer up to two decimals accuracy.

Answer (Numeric):

Accepted Answer : 1.3 to 1.4

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653388557

View Solutions (0)Total Mark : 3.00 | Type : SA

For what value of k , will the following function be continuous?

$$f(x) = \begin{cases} \frac{\tan 5x}{\sin 2x}, & \text{if } x \neq 0 \\ 2k & \text{if } x = 0 \end{cases}$$

Answer (Numeric):

Accepted Answer : 1.25

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 640653388559

View Solutions (0)Total Mark : 3.00 | Type : SA

For the matrix $S = \begin{bmatrix} 2 & -2 & 4 \\ 1 & 1 & -1 \\ -1 & 1 & -2 \\ -2 & -2 & 2 \end{bmatrix}$ what is the dimension of its column space?

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 640653388562

View Solutions (0)Total Mark : 3.00 | Type : SA

What is the length of orthogonal projection vector of $(4, 4, 6)^T$ on to a line through $(2, 3, 6)^T$? Round the answer to nearest integer.

Answer (Numeric):

Accepted Answer : 8

Your score : 0

Discussions (0)Suggest Correction

Question 13 : 640653388564

View Solutions (0)Total Mark : 3.00 | Type : SA

If $A = PDP^{-1}$ where $P = \begin{bmatrix} 1 & 2 \\ 1 & -5 \end{bmatrix}$ and $D = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix}$, then what is the summation of all the diagonal elements of A^4

Answer (Numeric):

Accepted Answer : 1297

Your score : 0

Discussions (0)Suggest Correction

Question 14 : 640653388566

View Solutions (0)Total Mark : 3.00 | Type : SA

Find the values of α and β for which the matrix $A = \begin{bmatrix} 4 & \alpha \\ 4 & \beta \end{bmatrix}$ has eigenvalues equal to -2 and -6. Enter your answer as $2 \times \alpha + \beta \times 2$

Answer (Numeric):

Accepted Answer : -42

Your score : 0

Discussions (1)Suggest Correction

Question 15 : 640653388555

View Solutions (0)Total Mark : 2.00 | Type : SA

Does the limit for the following function exist at $x = -1$?

$$f(x) = \frac{|x + 1|}{x}$$

(Provide 1 as answer for 'Yes' and 0 for 'No'.)

Answer (Numeric):

Accepted Answer : 0

Your score : 0

Discussions (0)Suggest Correction

Question 16 : 640653388561

View Solutions (0)Total Mark : 3.00 | Type : MCQ

In a linear regression problem $X\theta = y$, with features $X = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 3 & 1 & 5 \end{bmatrix}$
 $y = (16, 9, 18)^T$, what is the weight vector θ ?

Options :

$$(-3, 2, 5)^T$$

$$(3, 2, 5)^T$$

$$(-1, 1, 6)^T$$

$$(-3, 5, 2)^T$$

Your score : 0
